

EE272 Homework 8 Analog Design Report

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Inverter Sizing

- 1) A screenshot of your simulated circuit including the 3 inverters and supplies. (5)

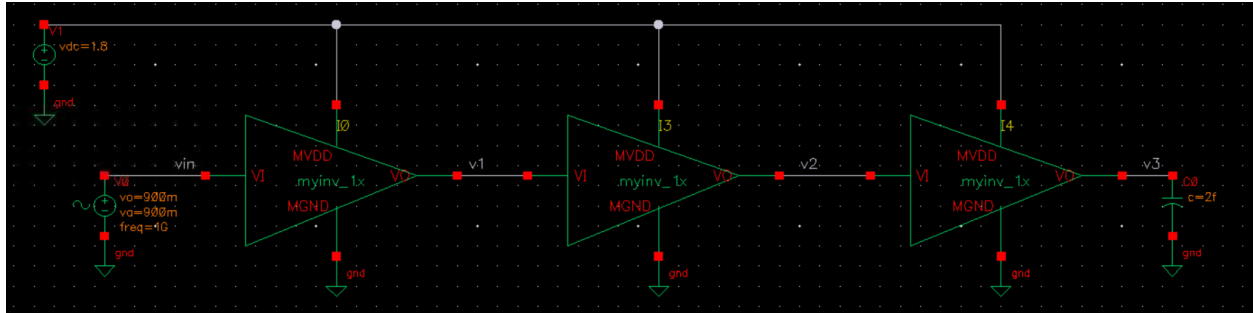


Figure 1. Schematic testbench: 3 inverters + supplies

- 2) A screenshot of the waveform including the sinusoidal input signal and the “v2” output node. (10)

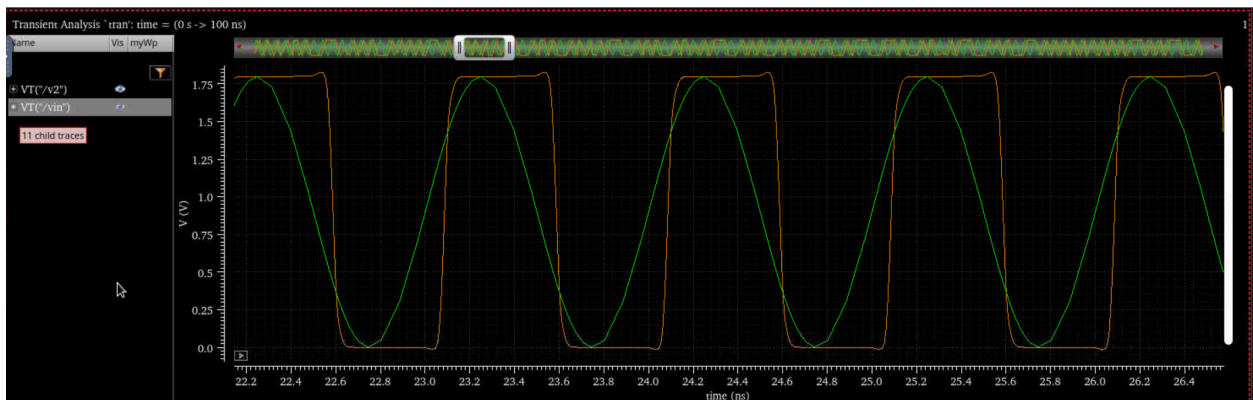


Figure 2. Waveform of v2 output and vin from 3 inverters and supplies

The transient simulation shows the 1 GHz sinusoidal input (vin) together with the output of the second inverter.

3) What is the optimal PMOS to NMOS ratio you found? (10)

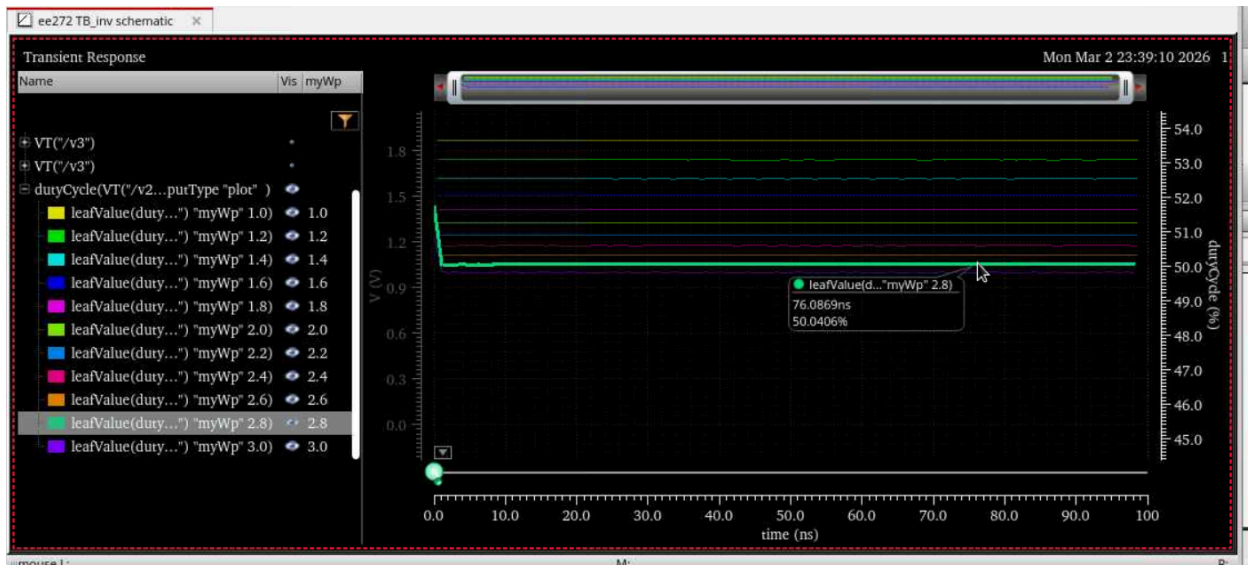


Figure 3. Transient sweep of PMOS sizing WP = 2.8 with DutyCycle of 50.0406%

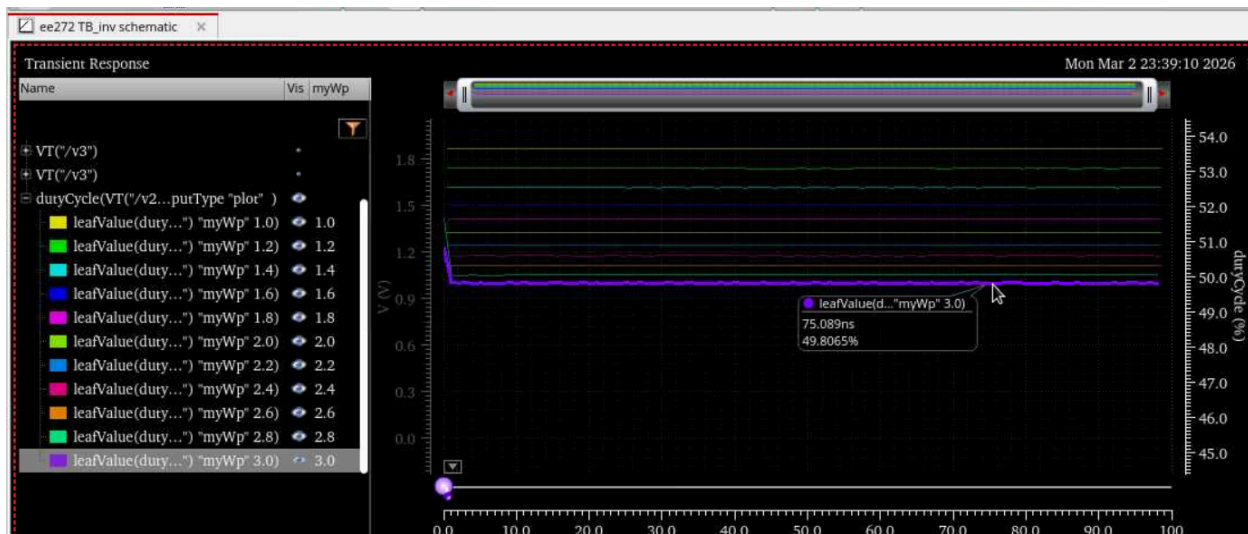


Figure 4. Transient sweep of PMOS sizing WP = 3.0 with DutyCycle of 49.81%

From the parametric sweep results, the PMOS width values closest to a 50% duty cycle were myWp = 2.8 and myWp = 3.0. The duty cycle at myWp = 2.8 was approximately 50.04%, while at myWp = 3.0 it was about 49.81%. Since myWp = 2.8 produced a value slightly closer to 50%, I selected this as the optimal sizing. Therefore, the optimal PMOS to NMOS ratio is approximately 2.8:1.

Inverter Layout

4) A screenshot of the finished inverter layout. (5)

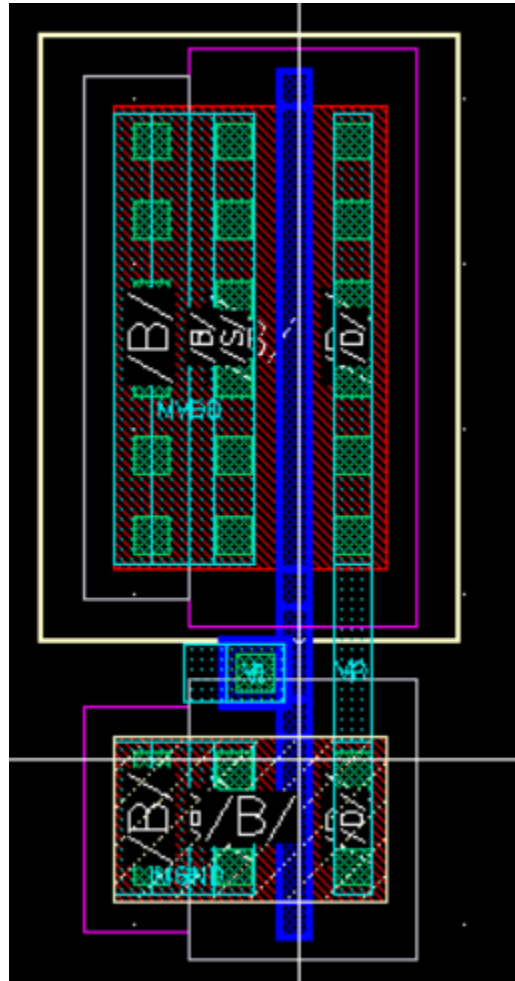


Figure 5. Screenshot of Inverter Layout with Wp of 2.8

- 5) A screenshot of the viewer after running DRC, showing only the remaining density errors. (10)

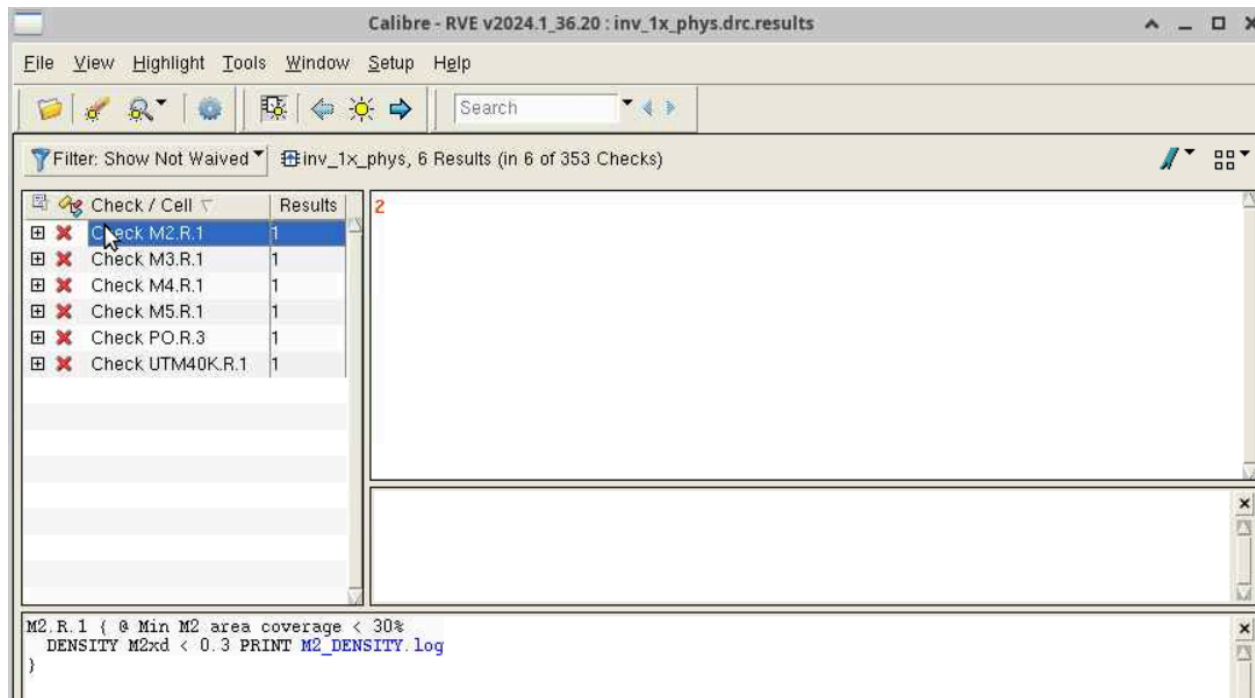


Figure 6. Screenshot of layout inverter design with just density violations.

6) A screenshot of the passing RVE viewer after running LVS. (10)

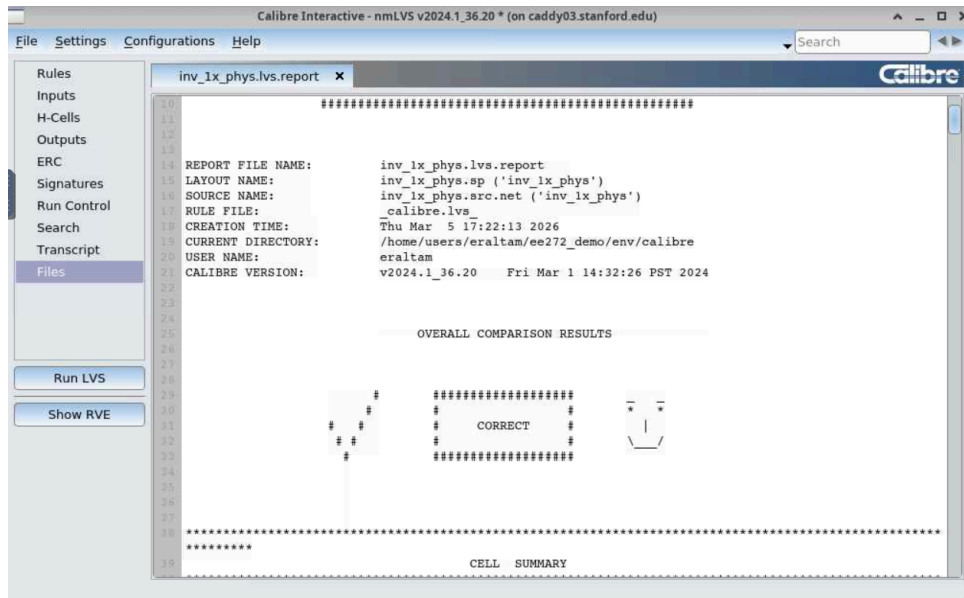


Figure 7. Screenshot of inverter LVS passing the compilation

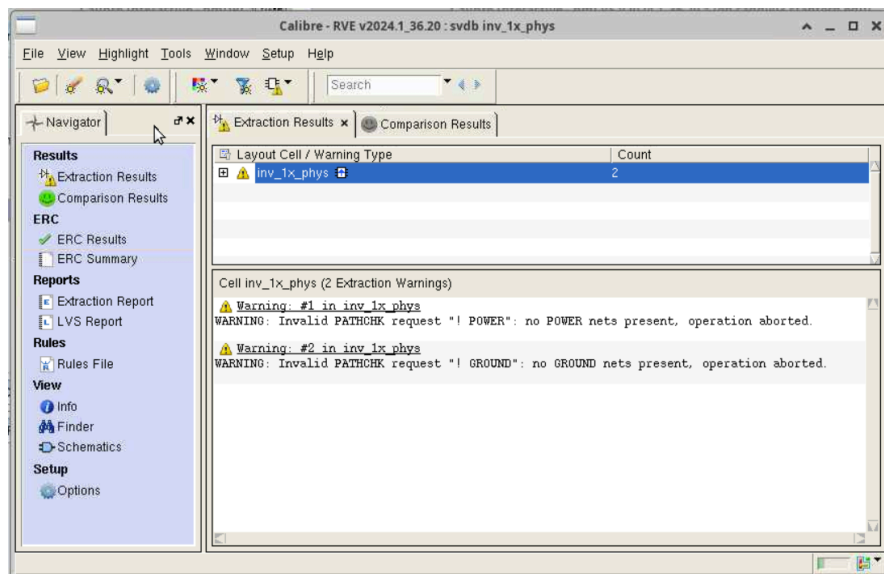


Figure 8. Screenshot of inverter RVE result

7) A screenshot of the Calibre view with the schematic viewer. (5)

The LVS results confirm that the layout netlist matches the schematic netlist. The Calibre comparison viewer shows the layout and schematic connectivity with no mismatches, indicating a clean LVS result.

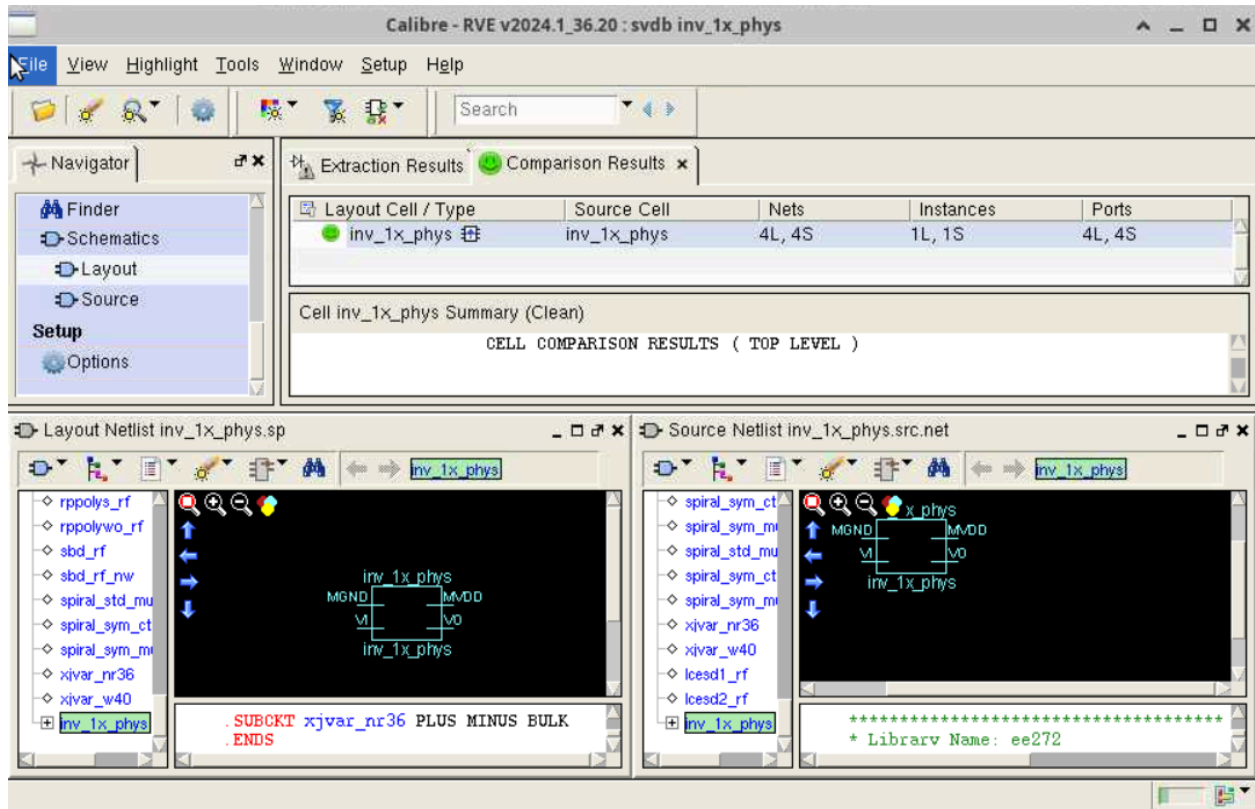


Figure 9. Screenshot of inverter calibre view with the schematic viewer.

Ring Oscillator

- 8) What is the oscillation frequency of your ring oscillator with schematic only? Include a screenshot of the waveform and/or whatever tools you used to calculate the frequency.
(10)

This setup uses the schematic-only simulation flow because the switch view list includes schematic views and does not call any extracted or post-layout view. Therefore, Spectre runs the ring oscillator from the schematic netlist only, so the measured oscillation frequency corresponds to the schematic-only oscillator.

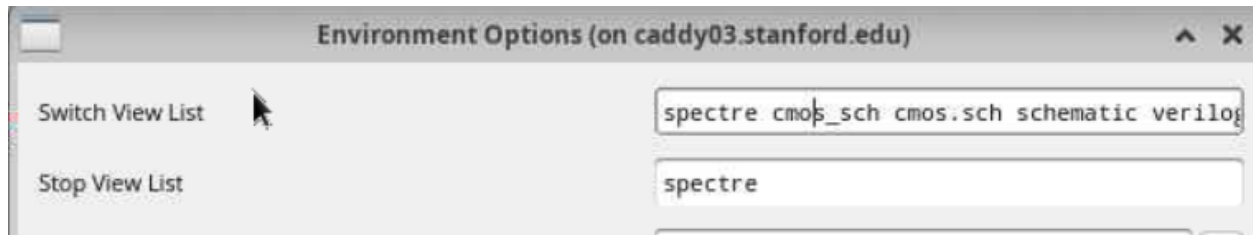


Figure 10. Environment Options showing that the simulation uses the schematic view list only

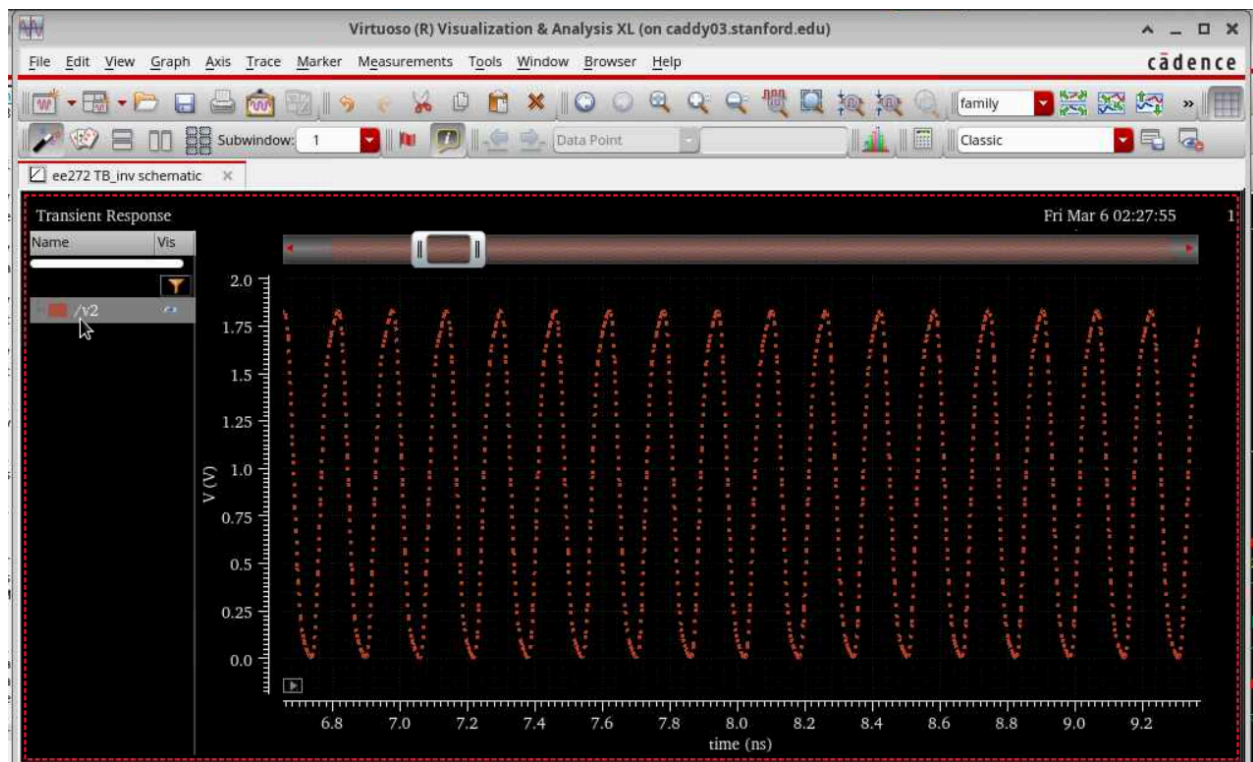


Figure 11. Simulation of v2 signal.

- 9) What is the oscillation frequency of your ring oscillator with the post-layout extracted inverters? Include a screenshot of the waveform and/or whatever tools you used to calculate the frequency. (10)

The ring oscillator was simulated using the post-layout extracted inverter views. In the Environment Options, the Switch View List prioritizes the extracted view (calibre) ahead of the schematic views, so Spectre uses the extracted netlist including layout parasitics. From the transient waveform, the oscillation frequency measured with the Cadence Calculator using $\text{freq}(\text{VT}("/\text{V2}"))$ at a 900 mV threshold is approximately 5.583 GHz.

As expected, the post-layout oscillation frequency is lower than the schematic-only result because the extracted view includes layout parasitics that increase the inverter delay.

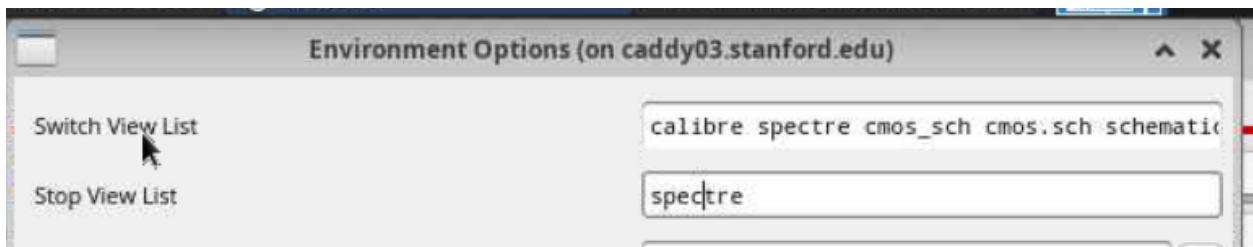


Figure 14. Environment Options showing calibre prioritized in the Switch View List, indicating that the simulation uses the post-layout extracted inverter view.

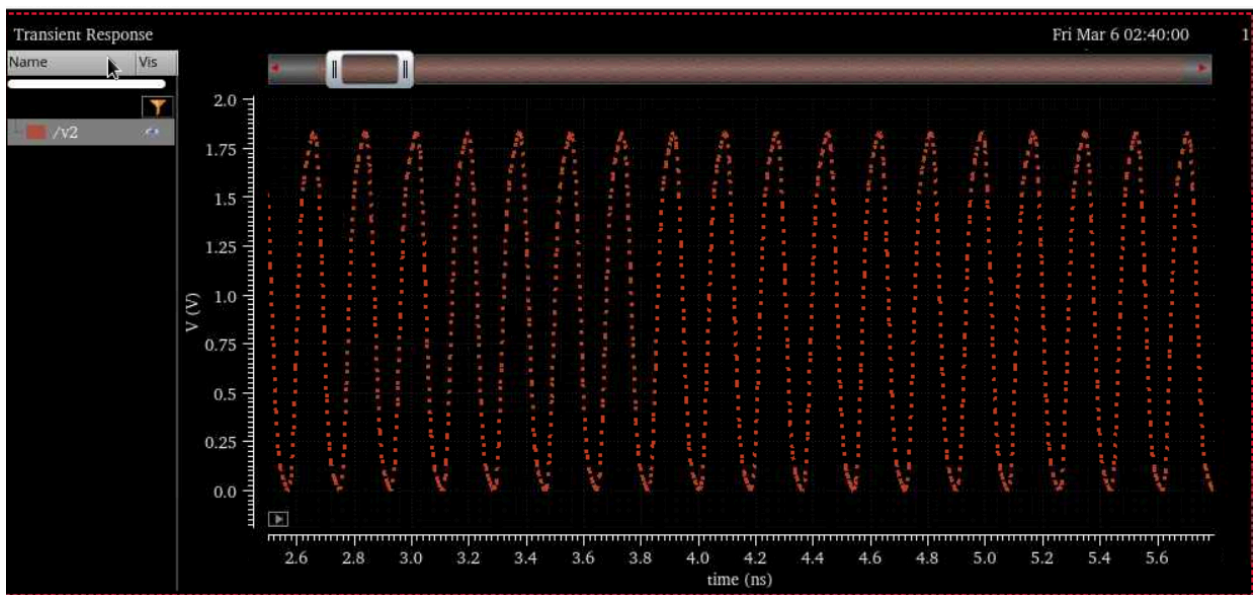


Figure 15. Simulation of v2 signal.

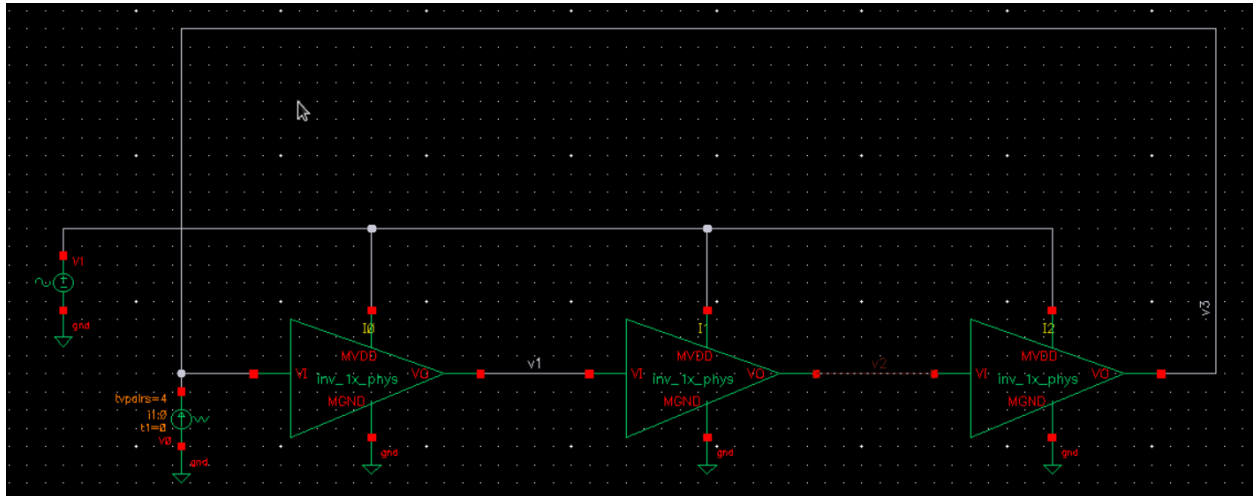


Figure 16. Schematic of this experiment

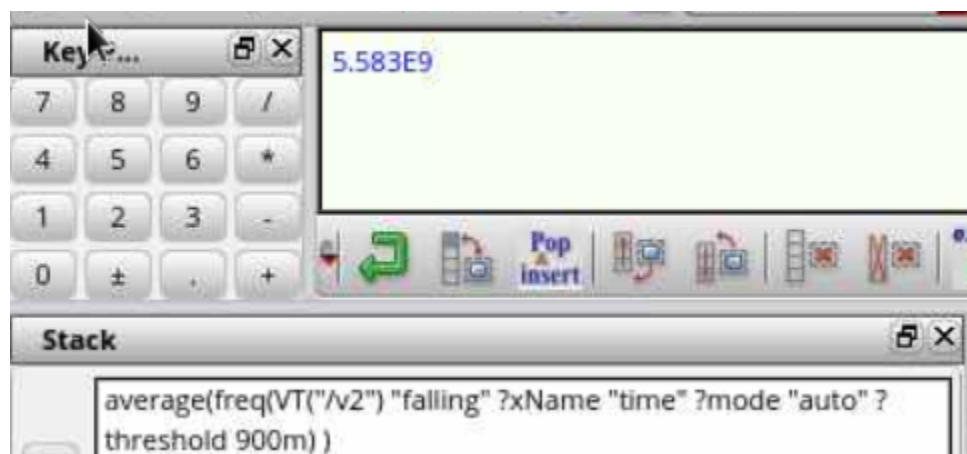


Figure 17. Cadence Calculator measurement of the oscillation frequency for the ring oscillator using the post-layout extracted inverter views.

- 10) A screenshot of the finished ring oscillator layout, including screenshots of passing DRC and LVS. (15)

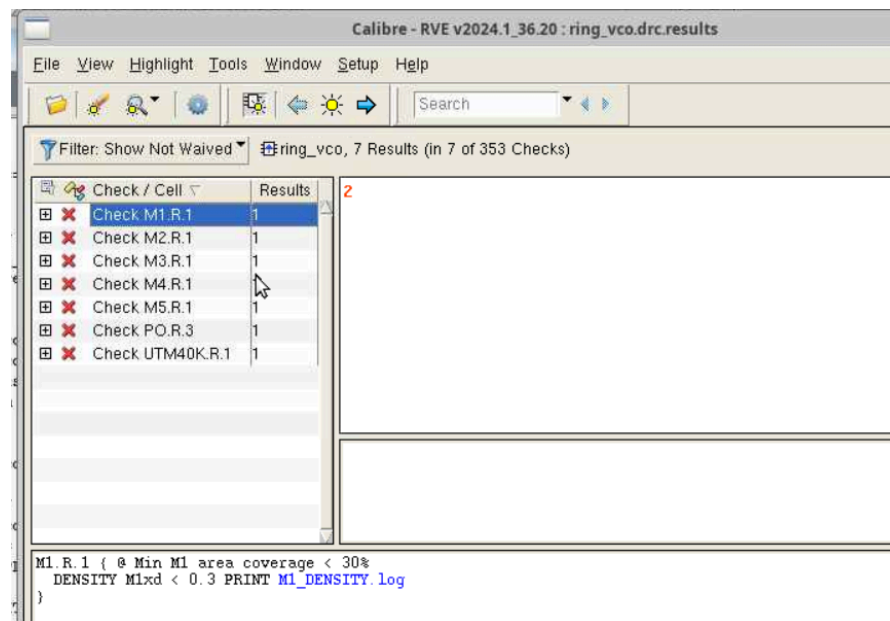


Figure 18. Screenshot of DRC with just density violations

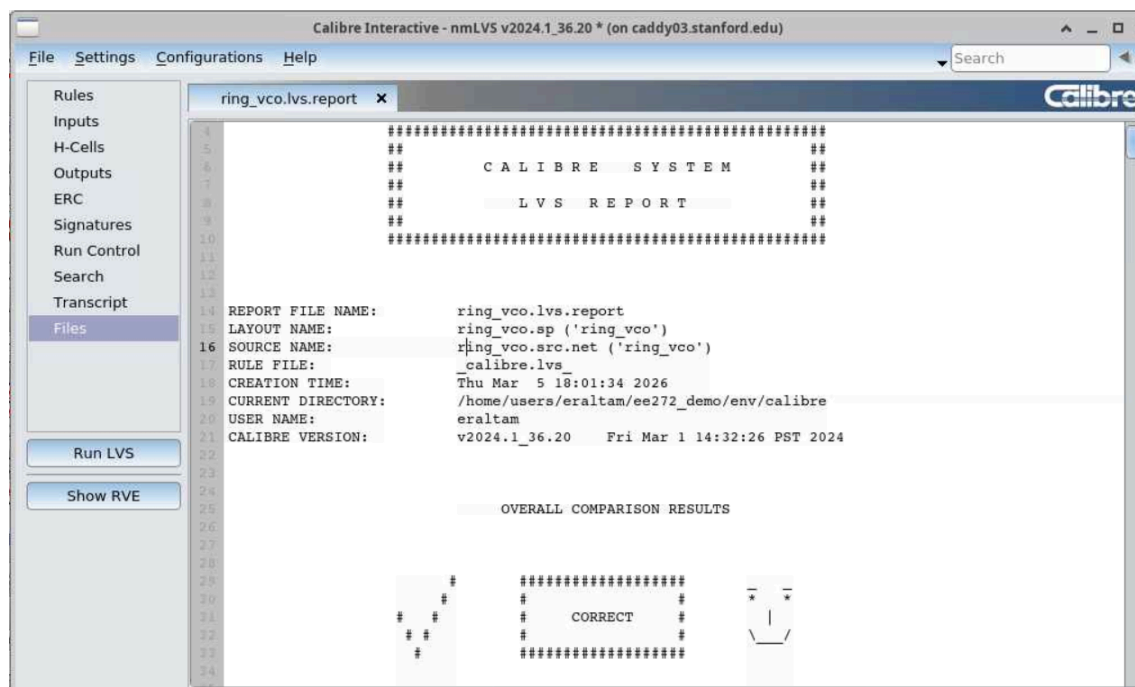


Figure 19. Screenshot of inverter LVS passing the compilation

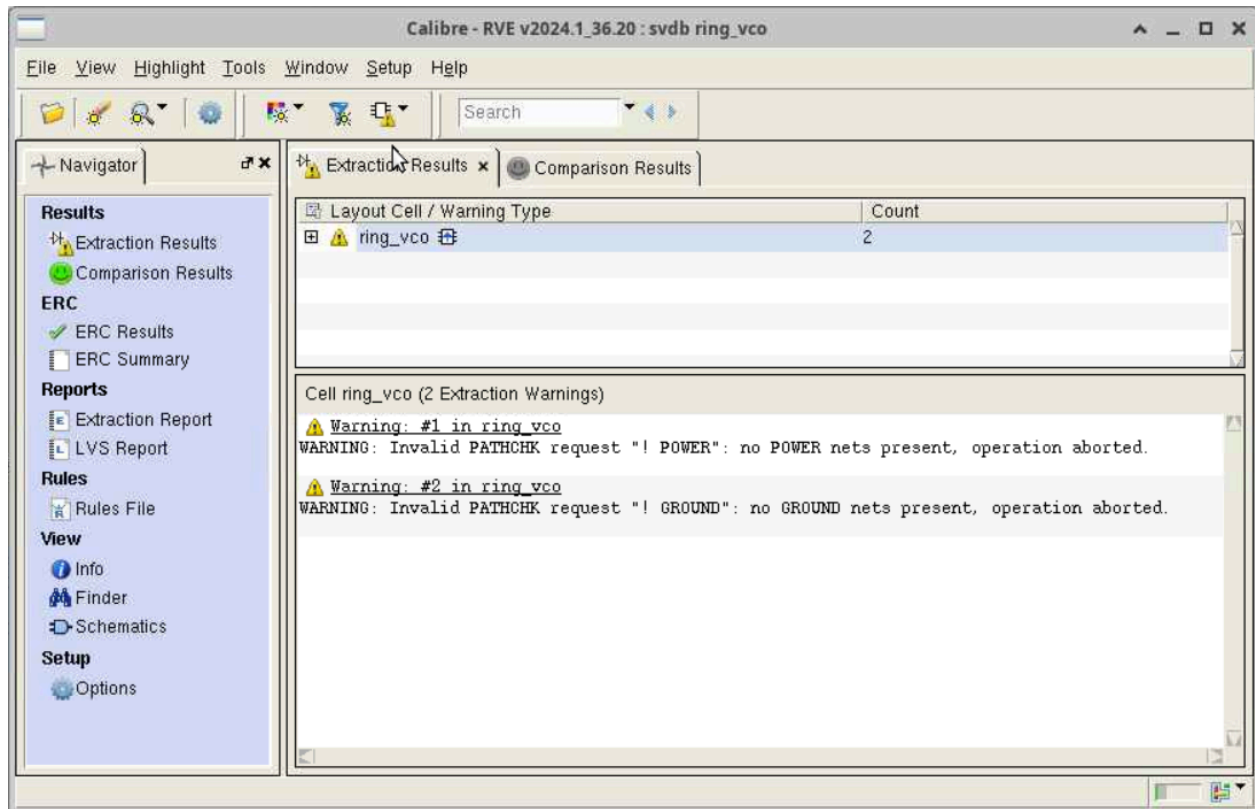


Figure 20. Screenshot of ring oscillator LVS passing the compilation

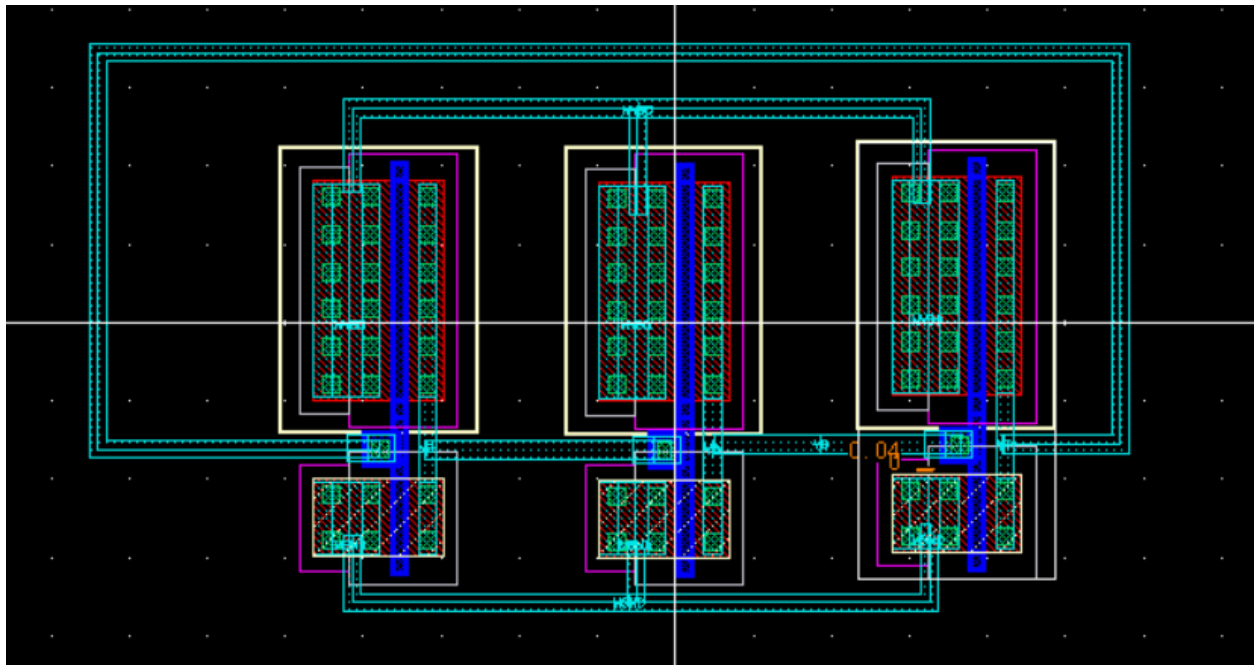


Figure 21. Screenshot of ring oscillator layout

- 11) What is the oscillation frequency of your ring oscillator with the full layout? Include a screenshot of the waveform and/or whatever tools you used to calculate the frequency. (10)

The ring oscillator was simulated using the full layout. In the Environment Options, the Switch View List prioritizes the extracted view (calibre) ahead of the schematic views, so Spectre uses the extracted netlist including layout parasitics. From the transient waveform, the oscillation frequency measured with the Cadence Calculator using `freq(VT("/VO"))` at a 900 mV threshold is approximately 5.307 GHz.

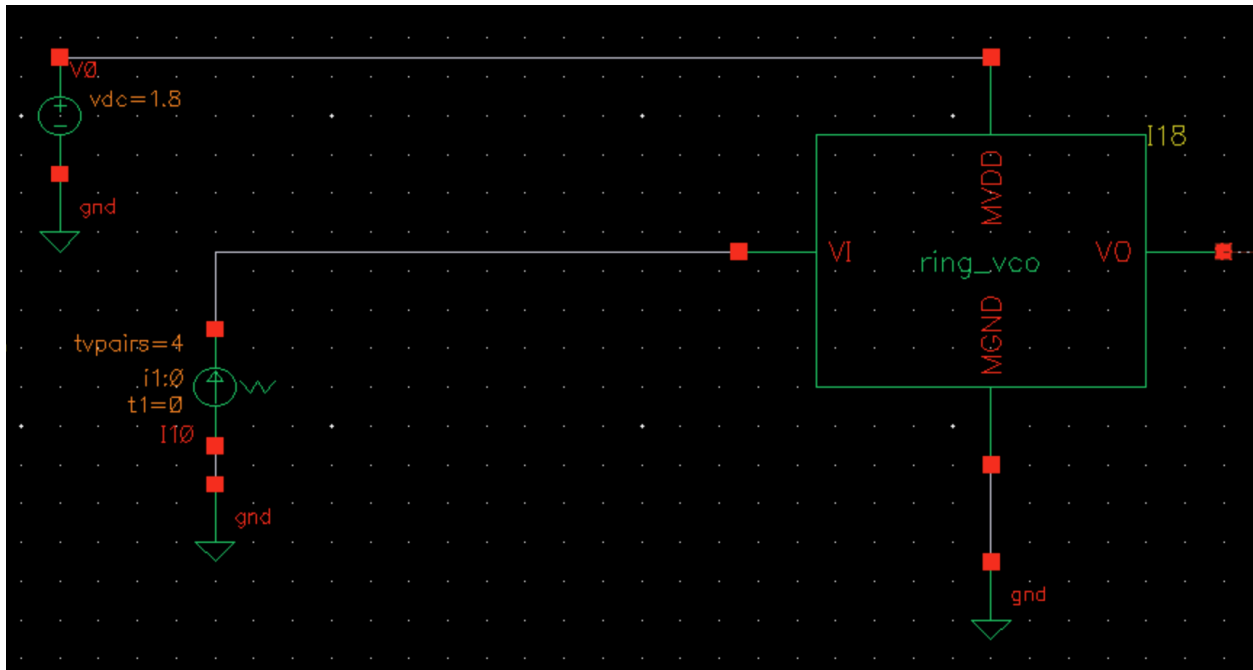


Figure 22. Schematic of this experiment

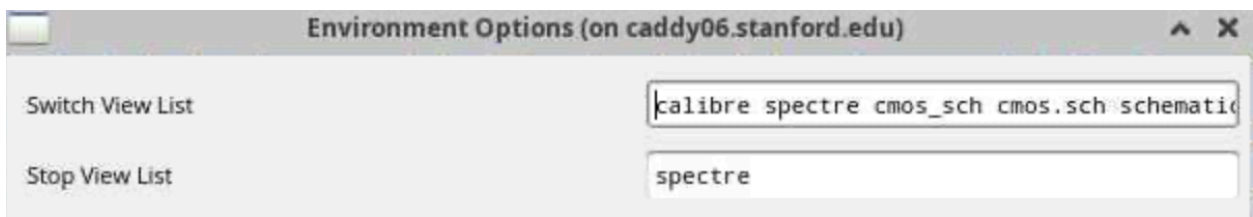


Figure 23. Environment Options showing calibre prioritized in the Switch View List, indicating that the simulation uses the post-layout ring oscillator.

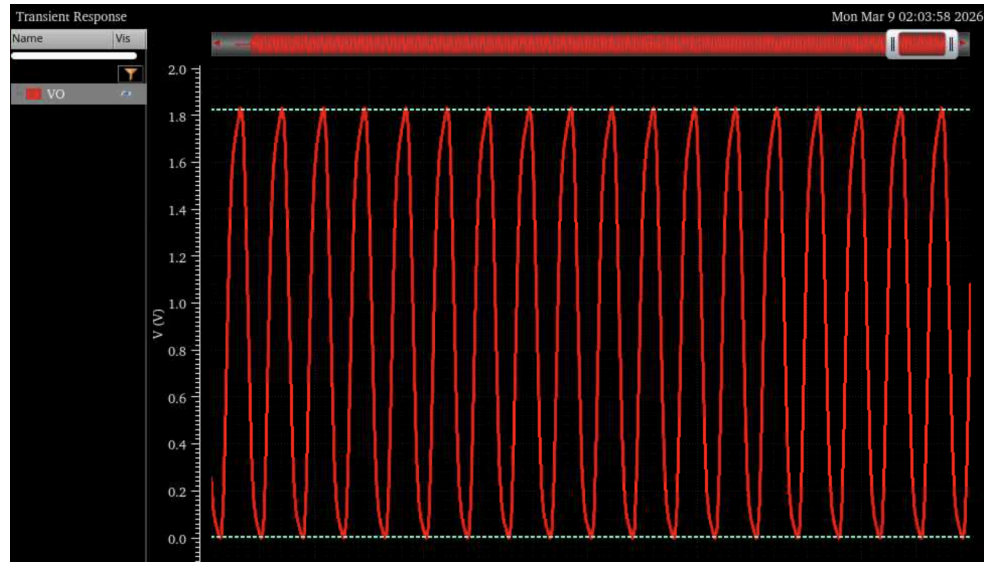
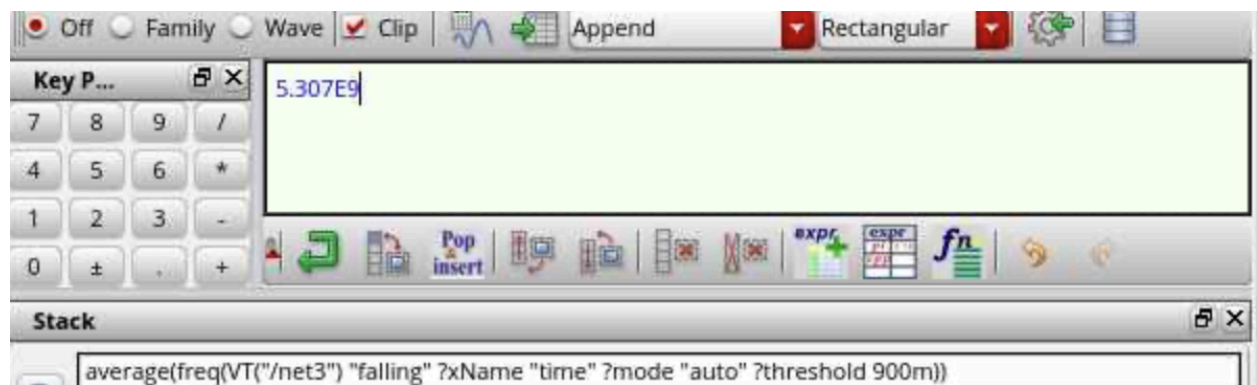


Figure 24. Simulation of VO signal



Cadence Calculator measurement of the oscillation frequency for the ring oscillator using the full layout view.